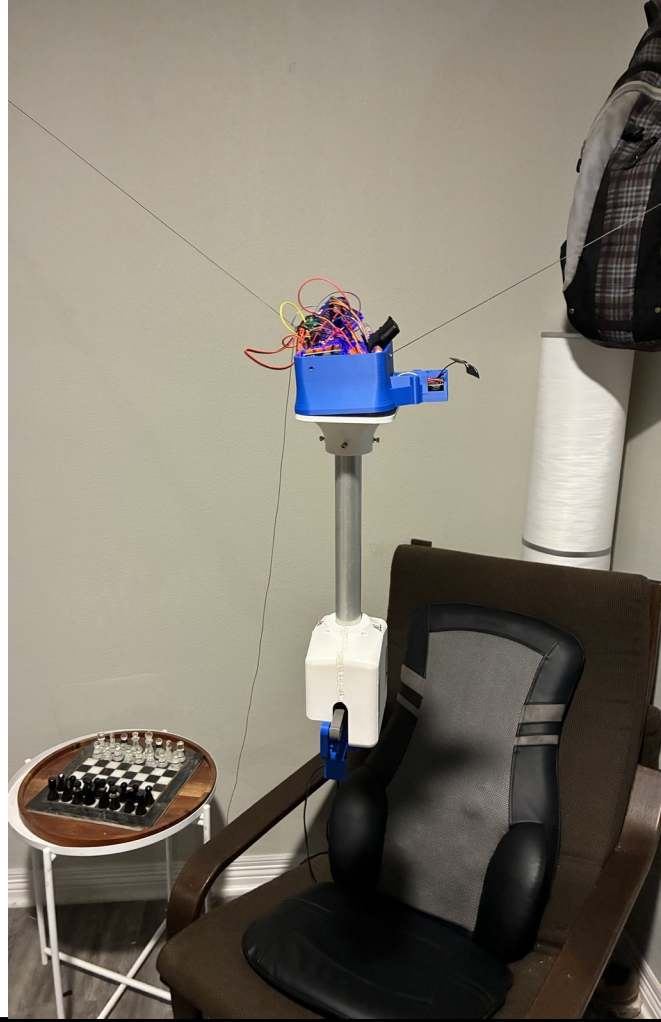
RoomCleaner

A four-cable ceiling robot that picks laundry off the floor

Jasper Buntinx | Mechanical Engineering | Texas A&M

October 27, 2026



About me

Education: Texas A&M

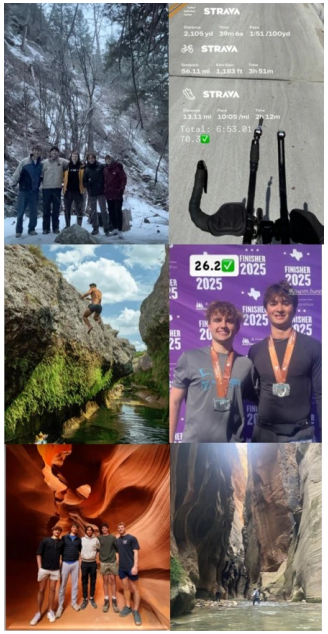
- Bachelor's, Mechanical Engineering, August 2027

Experience

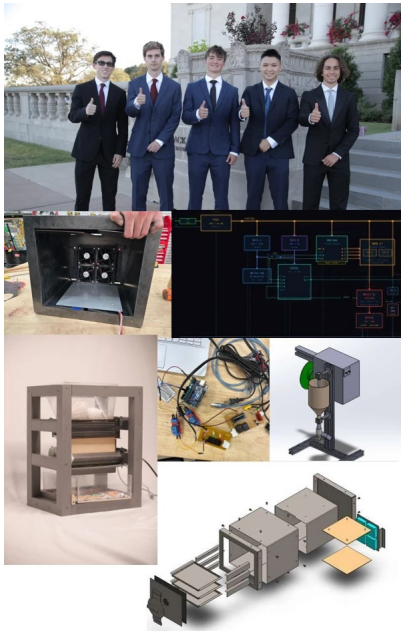
- ITP Lab, undergraduate research
- Rochester Sensors, mechanical intern
- Petrobras, mechanical intern
- Aggies Create, EcoFil electrical lead

Hobbies

- Personal projects
- Soccer
- Running
- Bouldering
- And man, do I love hiking.



Outdoors and endurance



Engineering



Community

Project scope: a robot that picks laundry off the floor

The problem

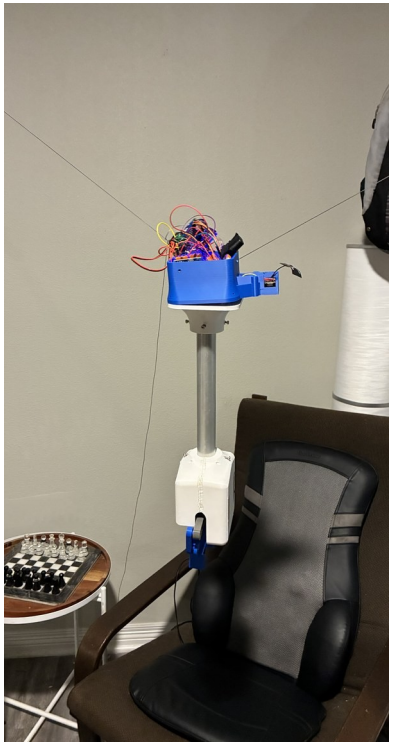
- Clothes end up on the floor every day; picking them up is a chore
- A floor robot fights the clutter it is meant to clear: furniture, rugs, cords
- A gantry would put a frame across the whole room
- Goal: any garment, anywhere on the floor, into the hamper, with nobody touching it

The proposed solution

- A cable robot: four winch boxes high on the walls hold one claw on four cables
- Changing the four cable lengths moves the claw anywhere in the room
- Nothing on the floor; only the claw moves: 0.60 kg (measured)
- Each box is self-contained: slides off the wall and unplugs when not needed

How it works

- The claw camera locates itself on a printed floor board and finds the garment
- Descend, grip, lift, carry, release over the hamper
- Each move approved on the phone; PAUSE holds; the power strip is the stop
- Success: one garment floor to hamper, no human touch
- Test path: 1 lb weight first (design), then jeans, 0.9 kg (assumed)



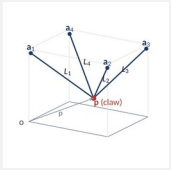
The claw on its cables, room corner

How cable length becomes position, and torque becomes pull

Cable length → position

$$L_i = \|\mathbf{a}_i - \mathbf{p}\|$$

$\mathbf{a}_i$ anchor, $\mathbf{p}$ claw



- where L_i: cable length, mount i to claw (m); a_i: cable exit point at mount i; p: claw position; r: spool radius incl. wound line (mm)
- Line per spool turn = 2π·r: 2π × 7.5 mm core (design) = 47.1 mm → 2π × 10 mm full spool (calc, 20 ft wound) = 62.8 mm (calc)
- Four cable lengths, three position unknowns: overdetermined, solved by least squares
- The camera on the floor board fixes the anchors and cable lengths at startup (calibration)
- Validated in the simulator and by held-out poses during calibration (sim); no physical position measurement yet

Torque → pull on the line

$$I = \frac{V_{\text{ref}}}{8\,R_s} \qquad \tau_{\text{hold}} = \tau_{\text{rated}}\,\frac{I}{I_{\text{rated}}}$$

$$\tau_{\text{run}} \approx 0.7\,\tau_{\text{hold}}\ \text{(assumed)} \qquad F = \frac{\tau}{r}$$

8: A4988 current-sense constant (datasheet)

- where I: phase current (A); Vref: driver reference voltage (V); R_s: current-sense resistor (Ω); τ_hold: holding torque at I (N·m); τ_rated, I_rated: listed holding torque and current; τ_run: running torque (N·m); F: line pull (N); r: spool radius (m)
- At 1.5 A (rated; chosen, design): τ_run = 0.7 (assumed) × 0.36 N·m = 0.252 N·m (1.5 A, 0.36 N·m: datasheet, SIMAX3D NEMA 17 42-38 seller listing); F = 0.252 / 0.0075 m = 34 N empty, 0.252 / 0.010 m = 25 N full (calc)
- Validated: Vref on the board (measured); 0.7 running fraction (assumed); pull test (planned)

Pull → what four cables hold

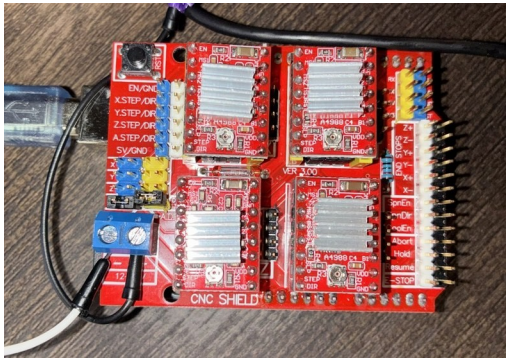
$$T = \frac{W}{4\sin\varphi} \qquad \sin\varphi = \frac{d}{\sqrt{d^2 + D_h^2}}$$

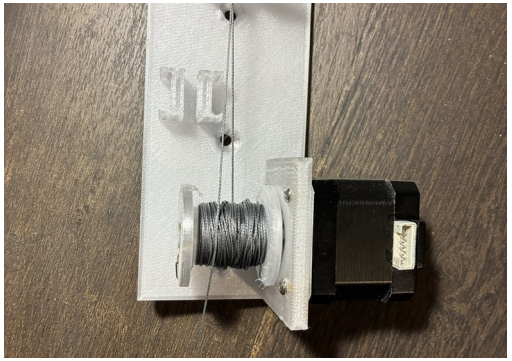
- where T: tension per cable (N); W: supported weight (N); φ: cable angle above horizontal; d: claw depth below the anchors (m); D_h: horizontal distance, claw to anchor (m)
- W = (0.454 kg test, design + 0.45 kg claw budget, design) × 9.81 (handbook) = 8.86 N (calc)
- D_h = √(1.676² + 1.372²) = 2.166 m (calc; half the 3.353 × 2.743 m room, measured)
- d = 0.15 m: 8.86 / (4 × 0.069) = 32 N; 16 / 8.3 / 4.6 N at d = 0.3 / 0.6 / 1.2 m (calc)
- Shallow cable: T ≈ W·D_h / (4d), so halving the depth doubles the tension
- Validated: calculated only; no cable tension has been measured on the rig

Generation 1: one central unit

The first design

- One Uno and one CNC shield drove four corner motors
- Wires from that one box to every corner
- An overhead ceiling camera scanned the room
- A printed corner plate screwed into a joist: a hook is a pivot, a plate is not
- Each cable found zero against a bead at the outlet
- Bench: a station lifted a test weight; the cable math held (measured)







The central board; a gen-1 corner plate; a gen-1 station lifting a test weight

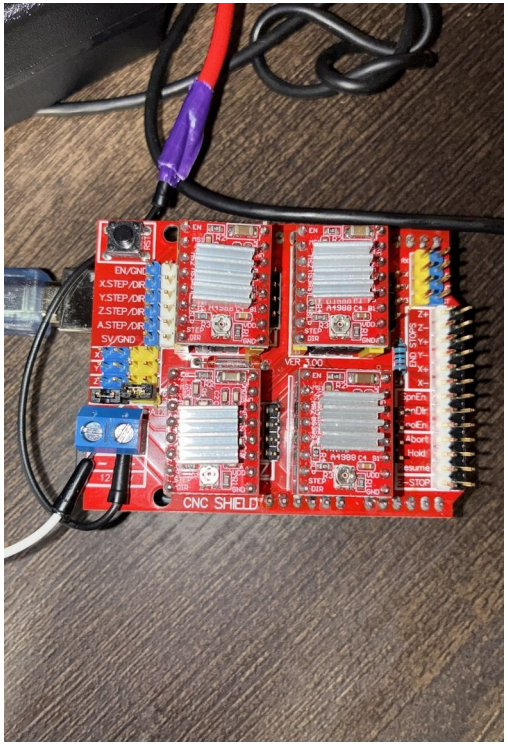
Problems with Generation 1

One central board

- Every motor and every sensor wired back to one place
- One board steps all four motors; a timing error skews the cables
- Skew $\Delta x = v \cdot \Delta t$ = 50 mm/s max cable speed (design) × 0.05 s timing error (assumed) = 2.5 mm (calc)
- where Δx: cable length mismatch between winches (mm); v: cable speed (mm/s); Δt: timing error between motors (s)
- Moving a winch means re-running wire to the box

Ceiling camera and bead homing

- The ceiling camera: one more device to mount, wire and calibrate per room
- From above, the claw and cables hide the grasp spot
- Its floor map assumes a flat floor; retired Oct 21
- Bead homing fixes a minimum length on every cable: next slide



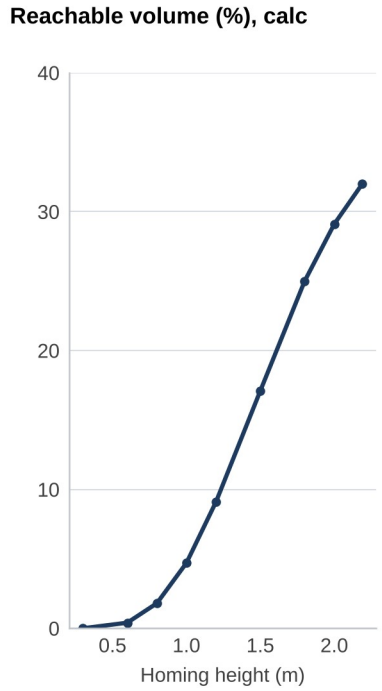
The central board: one shield, four drivers

Homing: one floor board instead of four beads

Bead homing

$$\text{reachable}(\mathbf{p}) \Leftrightarrow \|\mathbf{a}_i - \mathbf{p}\| \geq L_{i,\text{home}} \quad \text{for every } i$$

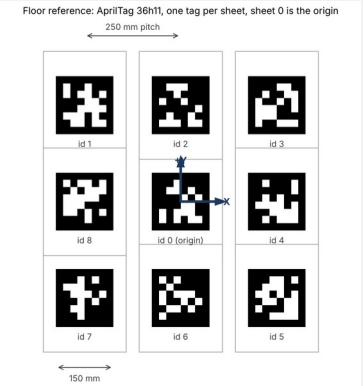
- where a_i: cable exit point at mount i; p: claw position; L_i,home: cable length at the homing pose (m)
- A stopper bead cannot pass its outlet: after homing, no cable can be shorter than at home
- Homing all four at once fixes a minimum length on every cable at the same time
- Reachable volume = grid count of the condition over the working volume (calc): 0.0 % (home 0.3 m) to 32.0 % (home 2.19 m); 4.7 % at 1.0 m, 29.1 % at 2.0 m; homing heights (design)



Grid count of the condition above

Floor-board homing

- The camera measures the claw pose against tag 0
- Cable lengths follow from L_i = ‖a_i − p‖
- where a_i: cable exit point at mount i; p: claw position; L_i: cable length from mount to claw (m)
- No length is clamped: the whole working volume stays available
- Nine sheets, 150 mm tags, 250 mm pitch, id 0 = origin (all design)



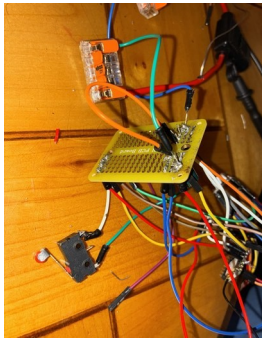
What a winch on a wall needs to work

The switch (button)

- Things go wrong: a camera fails, a spool overwinds, tension rips a mount out, a motor stalls
- A switch the cable's bead can hit: that station stops
- Fail-safe wiring: LOW at rest, HIGH pressed or with a broken wire

The encoder

- On the spool: measures the discrepancies while testing; data first
- Confirms every commanded step
- Catches slip: the tension sensor
- If the camera proves good enough, it can go later



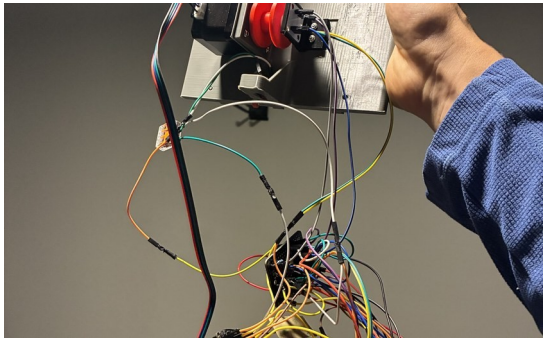
Homing switch



Switch pressed into the slide-in

And now all this wiring, from one box

- Motor, switch and encoder in every corner: 10 conductors per corner (design)
- Routes 6 + 21 + 23 + 32 ft (measured) = 82 ft; × 10 = 820 ft (250 m) of wire to one box (calc)



One slide-in, its harness hanging

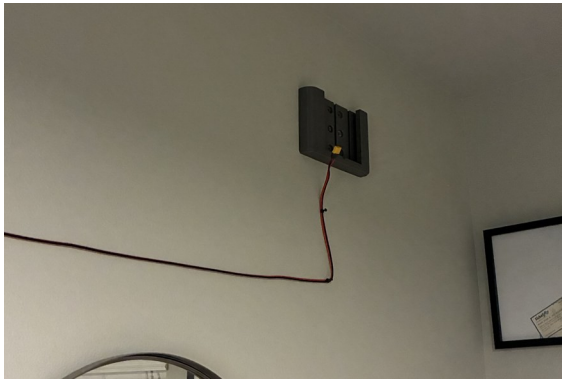
The solution: a modular box that only needs power

A box with its own brain

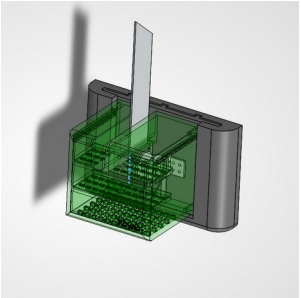
- An ESP32 in each box, with its driver, buck, switch and encoder
- The only thing going in is power: one 12 V pair (supply label, datasheet)
- 82 ft of routes (measured) × 2 conductors = 164 ft (50 m), not 820 ft (calc)
- Power can come from anywhere, so a box sits in any position and orientation
- The room's geometry no longer limits where the winches go

Wire drop: Gen 1 leads vs one power pair

- $V = I \cdot \rho \cdot 2L/A$; $\rho = 1.724 \times 10^{-8}$ Ω·m (handbook), L = 9.75 m / 32 ft one way (measured), I = 2 A (assumed)
- where V: voltage drop (V); I: current (A); ρ: copper resistivity (Ω·m); L: one-way run length (m); A: conductor area (m²); $R = \rho \cdot 2L/A$: loop resistance (Ω); $P = I^2 \cdot R$: power lost in the wire (W)
- 24 AWG Gen 1 motor leads (measured), A = 0.205 mm² (handbook): R = 1.64 Ω, 3.29 V (27 % of 12 V), 6.6 W (calc)
- 16 AWG power pair (design), A = 1.309 mm² (handbook): R = 0.257 Ω, 0.51 V (4.3 %), 1.0 W (calc)
- Thin motor leads over a room-length run lose a quarter of the supply; one thick power pair loses 4 % (calc, above)
- 16 AWG: $30 per 100 ft (assumed price) × 65 ft (design cut list) = $19.50 (calc)



The wall mount: one power plug



The station: board, driver, motor, fin (CAD)

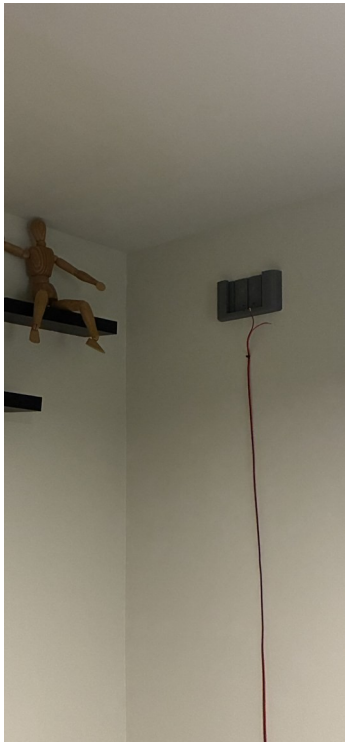
Modular, for the everyday user

Unmount, remount, plug in

- A box slides off its wall mount and unplugs
- Guests coming: each box goes in a drawer
- Back on: one plug, it joins the Wi-Fi by itself

No screwdriver on the wall

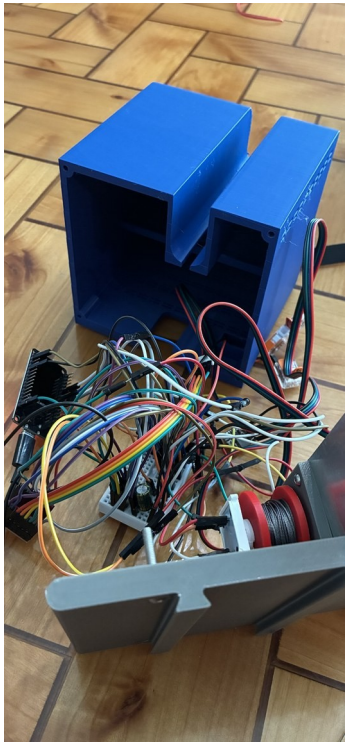
- Slip the station out of the mount, open the electronics
- Close it, slide it back: the mount and its screws stay in the stud
- The mount is separate: a stud today, a pole clamp later



The wall mount alone, lead hanging



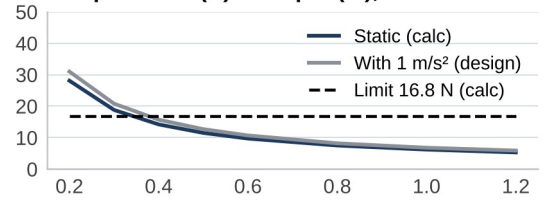
The station slid onto it



Shell open on the bench

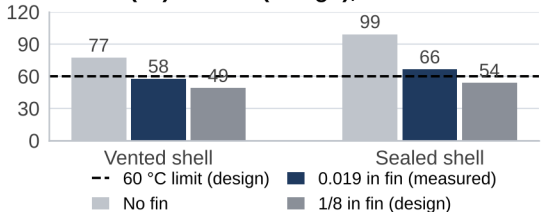
Trade-offs: what each component needs

Tension per cable (N) vs depth (m), calc



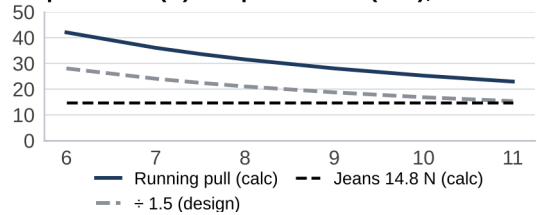
T = m(g + a)/(4 sin φ); m = 0.60 kg claw (measured) + 1 lb (design), a = 1 m/s² (design): d 0.4 m → 15.7 N < 16.8 N limit (calc); jeans 0.9 kg (assumed) need d ≥ 0.54 m (calc). where T: tension per cable; m: lifted mass; g: 9.81 m/s²; a: claw acceleration; φ: cable angle; d: depth below anchors

Motor case (°C) at 1.5 A (design), sim



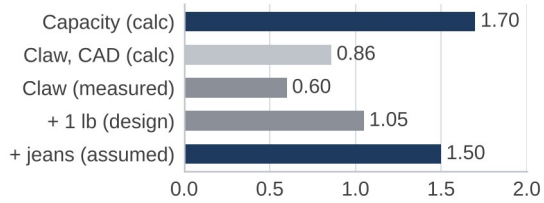
Need < 60 °C (design). Two-node thermal network (sim); P = I²·R = (1.5 A)² × 1.8 Ω = 4.05 W (calc; 1.5 A datasheet; 1.8 Ω research, conservative). where P: coil heat (W); I: phase current (A); R: coil resistance per phase (Ω)

Pull per winch (N) vs spool radius (mm), calc



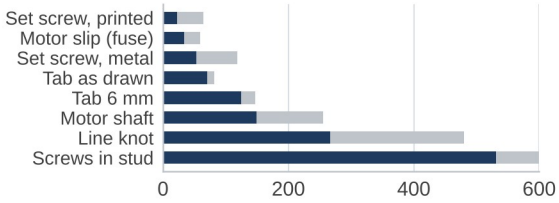
F = 0.7 (assumed) × 0.36 N·m (datasheet: SIMAX3D 42-38 listing) / r: 34 N at 7.5 mm (design), 25 N at 10 mm (calc); ÷ 1.5 skip margin (design) = 16.8 N ≥ 14.8 N jeans (calc). where F: line pull; τ: motor torque, 0.7 × holding when running; r: spool radius

Winches can hold vs must hold (kg)



k = 0.915 LP worst-cable factor (sim) × (1 + a/g) 1.102 = 1.008; 16.8 N / (1.008 × 9.81) = 1.70 kg vs 0.60 (measured) + 0.9 (assumed) = 1.50 kg: +13 % (calc); jeans 1.50 × 1.008 × 9.81 = 14.8 N. where k: worst-cable tension per unit m·g

Tension at which each part gives (N), calc



Estimates from hand calcs and published strengths (handbook); no stress concentrations. Motor slip: 0.36 N·m (datasheet) / 7.5 mm (design) = 48 N (calc)

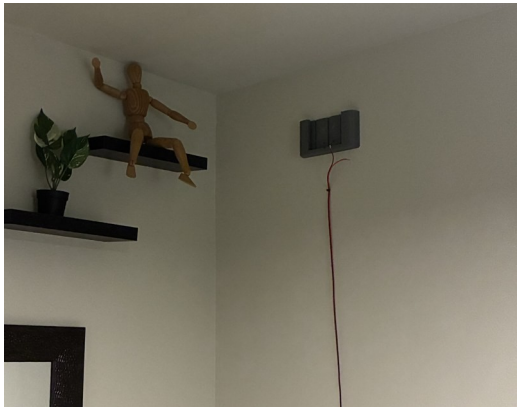
The factors

- Pull ∝ I; heat P = I²R ∝ I²
- 1.0 A earlier setting (design) → 1.5 A: pull × 1.5, heat × 1.5² = 2.25 (calc)
- Every gram on the claw is a gram of payload
- The printed set screw may slip before the motor: metal insert

The mount assembly: slide-in on a dovetail

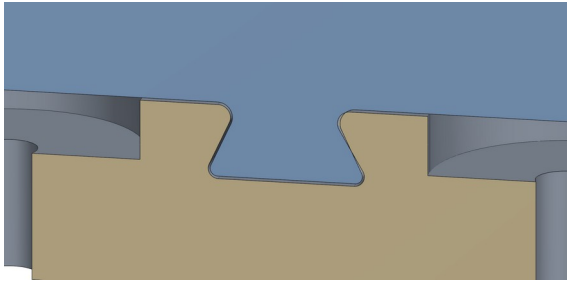
Slide-in mount and load path

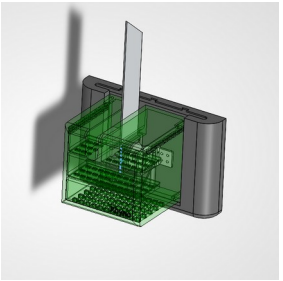
- The wall mount screws into a stud; the station slides onto it on a dovetail
- Dovetail: wall mount offset outward 0.175 mm per side (design); clearance set from fit coupons (measured)
- Load path: cable, spool, set screw, motor, tab, mount, screws, stud
- Motor slips first: $F = \tau / r = 0.36\ \text{N·m}$ (datasheet: SIMAX3D NEMA 17 42-38 listing) / 7.5 mm (design) = 48 N (calc)
- where F: cable tension at the spool (N); τ: motor holding torque (N·m); r: spool core radius (m)
- The spool encoder sees the slip: tracking fault, every station holds





The mount alone, before the station slides in; the station on the mount





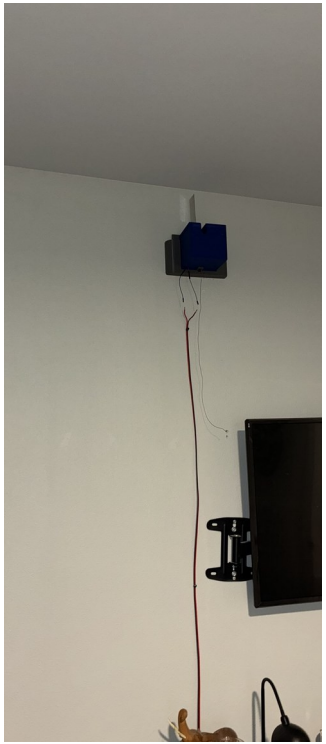
Rib in the dovetail groove (CAD); the station with its shell transparent (CAD)

The mount assembly: fin, vents and the wall

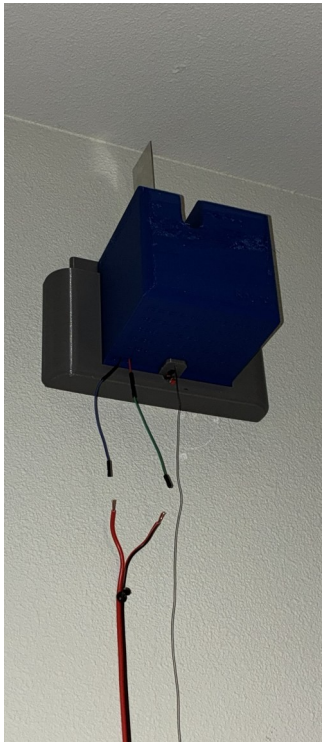
Fin, vents, stud

$$q_{\text{rad}} = \varepsilon \sigma A (T_s^4 - T_\infty^4), \quad \text{i.e.} \quad h_{\text{rad}} = \varepsilon \sigma (T_s + T_\infty)(T_s^2 + T_\infty^2)$$

- where q_rad: radiated heat (W); ε: emissivity; σ: Stefan–Boltzmann constant (W/m²K⁴); A: fin area (m²); Ts: fin surface temperature (K); T∞: room temperature (K); h_rad: radiation coefficient (W/m²K); h: convection coefficient (W/m²K); ΔT = Ts − T∞ (K); L: plate height (m)
- The fin: aluminium 6061 (design) on the motor face, out through a slot in the top of the shell
- Motor heat: P = I²·R = (1.5 A)² × 1.8 Ω = 4.05 W (calc; 1.5 A rated, datasheet: SIMAX3D NEMA 17 42-38 seller listing; 1.8 Ω per phase, research: comparable 42×38 NEMA 17 datasheets list 1.5–1.65 Ω, 1.8 used as conservative)
- where P: coil heat (W); I: phase current (A); I_peak: peak phase current (A); R: coil resistance per phase (Ω)
- With microstepping the two phase currents follow sin and cos, so total heat = I_peak²·R
- That heat leaves through the vents and the fin
- At Ts 50 °C, T∞ 23 °C: h_rad = 6.75·ε W/m²K (calc; σ = 5.67×10⁻⁸ W/m²K⁴, handbook; temperatures, design)
- Bare aluminium ε ≈ 0.05–0.09, black paint 0.92–0.96 (handbook)
- Natural convection: h = 1.42 (ΔT/L)^0.25 = 1.42 × (27 K / 0.2 m)^0.25 = 4.8 W/m²K on a 0.2 m vertical plate (calc, handbook correlation)
- Black paint: total 4.8 + 0.07 × 6.75 = 5.3 → 4.8 + 0.9 × 6.75 = 10.9 W/m²K, about 2× the heat per degree (calc); the motor contact square stays bare



On the TV wall



Fin through the slot

Manufacturing: in-house processes and fit control

Processes

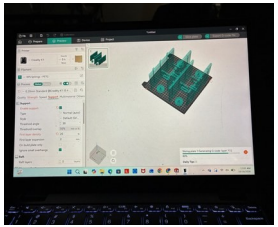
- PETG, FDM: 0.4 mm nozzle, 0.20 mm layers, supports at 30° (all design); loads along the layers
- Aluminium fin: sheared blank, five holes drilled in one mill setup
- First hole is the mill zero: pattern ±0.1 mm vs ±0.5 mm sheared edges (design tolerances)
- Heat-set inserts: M3 gear housing, M2.5 encoder board

Make vs buy

- Bought: steppers, drivers, boards, servo, camera, battery, encoders
- Made: every printed part and the fin
- Filament per station: 0.762 kg (calc: CAD volume × 1.27 g/cm³, handbook × fill, assumed) × $15/kg (assumed) = $11.43 (calc)
- Robot, three stations: $210.78 parts (listing prices, partly assumed) + 2.52 kg × $15/kg = $37.74 filament (calc)

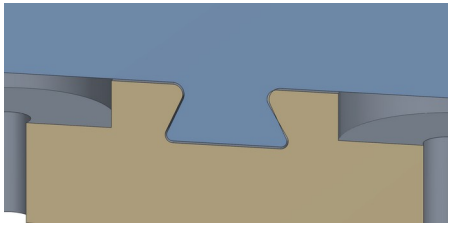
Fits

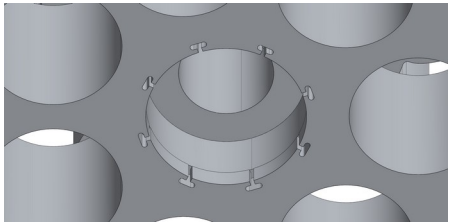
- Printed holes come out undersize: every printed fit carries an allowance
- Spool bore: Ø5.2 (design) printed too tight (measured) on the Ø5.0 shaft (datasheet); opened to Ø5.45 (design)
- Dovetail: wall mount offset outward 0.175 mm per side, 30° flanks (design)
- Bearing pocket Ø14.075 behind a Ø14.000 entry lip, 8 T-slots (design)
- Fit coupons printed first; the allowances above came from them (measured)
- A failed layer was kept on one shell only: the shell carries no load
- TPU 95A (datasheet) grip pads: next iteration





Slicer (K1, PETG); drilling the fin





Dovetail rib; bearing pocket and lip (CAD)

The claw, and why it is modular

The claw today

- Four-bar hands on a geared crank; 53 mm opening (design)
- One 7.4 V two-cell pack (datasheet): 6 V regulator for the servo, 5 V buck for the board (design setpoints)
- Weighed 0.60 kg with battery (measured) against a 0.45 kg budget (design)
- Grip per jaw, jeans: $N = m(g + a)/(2\mu) = 0.9 \times (9.81 + 1)/(2 \times 0.3) = 16.2$ N; 8.1 N at μ 0.6 (calc; 0.9 kg assumed; a = 1 m/s² design; μ 0.3 worst case, from published ranges)
- where N: normal force per jaw (N); m: garment mass (kg); g: 9.81 m/s²; a: claw acceleration (m/s²); μ: cloth-on-jaw friction coefficient
- Published fabric friction: 0.22–0.38 garment fabrics, 0.52–0.62 special fabrics, denim on skin 0.14–0.44; none is fabric on PETG, so 0.3 is the design worst case (handbook / literature)

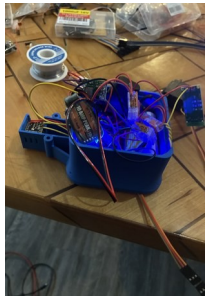
Modular, on purpose

- The gripper attaches at the tube, so other grippers can be tried
- A hard pinch was expected to grip better; testing showed alignment problems
- The first rack-and-pinion jaws left a gap on thin cloth (Sep 30): closed width 77.4 mm − 2 × pinion radius 18 mm × travel 2.09 rad = 2.0 mm (calc; CAD dimensions, design)
- TPU grip pads go on in the next iteration
- The v6 clamp: 2,061 g in CAD at solid density (calc) / 450 g budget (design) = 4.6 (calc); mass tracked per part since









Assembled; the hands (CAD); the v6 clamp (CAD); the housing open

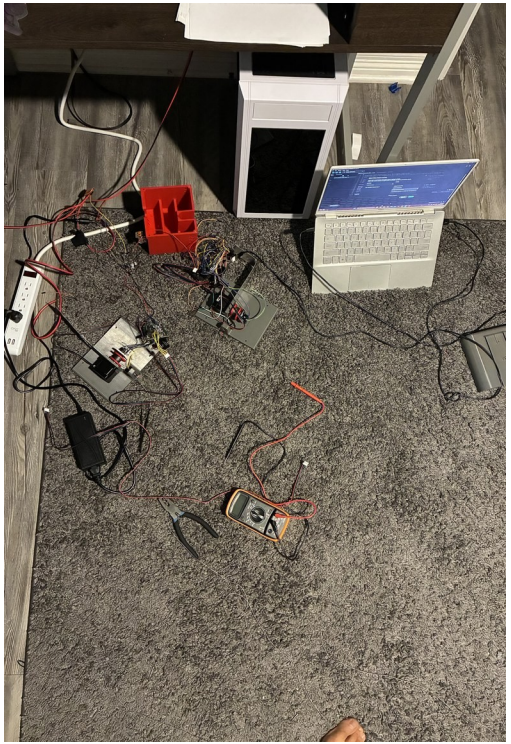
Electrical and testing: bench, then wall

Bench

- Station boards: bench checklist passed on both controller boards (measured)
- Claw: camera live view, battery sense and servo pass (measured)

On the wall

- A boxed station boots and joins Wi-Fi by itself
- Firmware updates over Wi-Fi; no USB cable after boxing
- Motors stay off until the phone arms them; updates only while disarmed
- Planned: pull test, hold test, first three-station lift



Bench: two stations, laptop, meter, 12 V supply (datasheet)



The first station on the wall

Startup and self-calibration: six states before the robot is READY

1 SELF_CHECK

- Claw online, battery above minimum
- Camera confirms the claw upright at the known height on the floor board
- Stations healthy, homing switches closed; arm

2 SLACK_TAKEUP

- One cable at a time reeled in until the camera sees the claw slide or tip (onset)
- Records spool position and slide bearing; backs off
- Bearings checked against the anchor layout; first fit of cable lengths

3 INITIAL_LIFT

- Pretension all cables; lift in small packets
- Abort on lateral drift, tilt, or commanded lift exceeding observed rise (slack cable)

4 VISUAL_INITIALIZATION

- Settled frame batch of the floor board
- Refit cable lengths and the room-to-marker transform

5 CALIBRATION

- Local → intermediate → expanded pose sets chosen by D-optimal design; refit between stages
- Held-out witness poses
- Accept on witness residuals, identifiability, predicted length sigma over the work volume

6 REFERENCE_VERIFICATION

- Optional bead references
- Trust level VERIFIED or LOCAL_ONLY from the margin check

No automatic retry; any station reboot forces the full sequence again.

Finding a mount from small moves, worked through

1. The equation, one per move

$$\|\mathbf{a} - \mathbf{p}_k\| - \|\mathbf{a} - \mathbf{p}_0\| = \Delta L_k$$

$$\Delta L_k = n_k \cdot \frac{2\pi r}{N}$$

- where a: mount position (unknown, m); p_0: claw at home, p_k: claw after move k (camera, m); ΔL_k: cable length change (m); n_k: microsteps counted; r: spool radius 7.5 mm (design); N: 3200 microsteps/rev (design)
- The change in distance from the mount to the claw must equal the change in cable length
- One microstep = 2π × 7.5 mm / 3200 = 0.01473 mm (calc)
- No absolute length needed: only changes

2. One move, checked against the guess

- Home p_0 = (1.68, 1.37, 1.00) m; move +100 mm in x → p_1 = (1.78, 1.37, 1.00) m (camera)
- Spool counts n_1 = 4283 microsteps → ΔL_1 = 63.07 mm (calc)
- Tape-measure guess a^o = (0.20, 0.15, 2.45) m, 228 mm from the true mount (illustrative)
- Predicted: $\|a^o - p_1\| - \|a^o - p_0\|$ = 2.4672 − 2.4044 m = 62.81 mm (calc)
- Residual r_1 = 62.81 − 63.07 = −0.26 mm: the guess is wrong
- Six moves (±x, ±y, +z, diagonal) give six residuals

3. Correct the guess, repeat

$$r_k = (\|\mathbf{a} - \mathbf{p}_k\| - \|\mathbf{a} - \mathbf{p}_0\|) - \Delta L_k$$

$$\mathbf{J}_k = \frac{\mathbf{a} - \mathbf{p}_k}{\|\mathbf{a} - \mathbf{p}_k\|} - \frac{\mathbf{a} - \mathbf{p}_0}{\|\mathbf{a} - \mathbf{p}_0\|}$$

$$\Delta\mathbf{a} = -(\mathbf{J}^T\mathbf{J})^{-1}\mathbf{J}^T\mathbf{r}$$

- where r_k: residual of move k (m); J_k: how r_k changes as a moves (unit vectors from claw to mount); Δa: correction (m)
- Gauss–Newton: Δa = (−136, −91, +127) mm (calc)
- Mount error 228 → 21 → 1.4 mm in two iterations (calc)
- Illustrative: perfect camera, one mount; the robot solves all mounts and length offsets together, weights camera noise, and grows the moves from local to expanded

Next iteration: what was not working, and what changes

3:1 printed gear stage

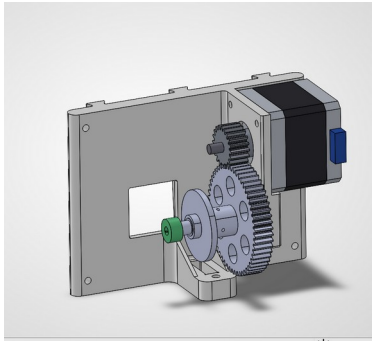
- 18T / 54T spur, module 1, 20° (design); printed PETG pinion; checked in CAD, pinion printed, not fitted
- Pull × 3 (design) × 0.9 η (assumed, conservative) = 2.7 at equal current (calc); at 1.0 A (design): heat × $(1.0/1.5)^2$ = 0.44 (calc)
- Mesh loss (Niemann): $H_v = \pi(u+1)/(z1 \cdot u) \cdot (1 - \varepsilon\alpha + \varepsilon1^2 + \varepsilon2^2) = 0.167$; $\eta = 1 - \mu \cdot H_v = 0.92–0.97$ for μ 0.2–0.5 (calc)
- z1 18, z2 54, u 3 (design); ε1 0.765, ε2 0.884 (calc); μ 0.2–0.5 dry polymer (assumed); 0.9 leaves room for bearing and print-roughness losses
- where η: efficiency (mesh, or stage); H_v: tooth loss factor; u = z2/z1: gear ratio; z1, z2: pinion and gear teeth; εα = ε1 + ε2: contact ratio; ε1, ε2: approach and recess contact ratios; μ: tooth friction coefficient

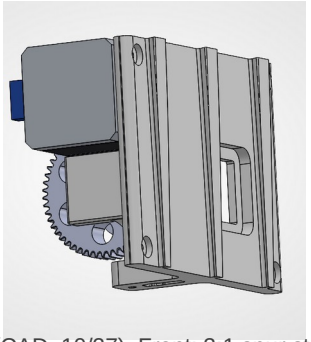
1° dovetail draft and a lighter back mount

- Draft 1° (design): the slide-in binds today; thicker at the top, it self-locates (planned)
- Back mount: 597 cm^3 solid (design, CAD) → about 200 cm^3, thinner walls (planned)

Also next / still planned

- Pull test at 1.5 A (design): the measured limit goes into the planner (planned)
- Hold test: 90 minutes at 1.5 A, motor below 60 °C (planned; 60 °C limit, design)
- First three-station lift: self-check, calibration, a short move with 1 lb (planned; 1 lb, design)





Next iteration (CAD, 10/27). Front: 3:1 spur stage, spool on its own shaft, encoder magnet cap. Back: slide-in rails, window, screw holes

Thank you

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